Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2016 – Paper 2 – Solutions**

June 2016 – Question 1 – Solution

(a) Let $f(x) = 2x^3 - x^2 + px + q$.

(i) Given that $x + 3$ is a factor of $f(x)$ and that there is a remainder of 10, when $f(x)$ is divided by $x + 1$, show that $p = -25$ and $q = -12$. [7]

Let $f(x) = 2x^3 - x^2 + px + q$.

Since $(x + 3)$ is a factor of $f(x)$, then by the factor theorem, $f(-3) = 0$.

$f(-3) = 2(-3)^3 - (-3)^2 + p(-3) + q$

$0 = -54 - 9 - 3p + q$

$0 = -63 - 3p + q$

$q = 3p + 63$ → Equation 1

Since there is a remainder of 10 when $f(x)$ is divided by $(x + 1)$, then by the remainder theorem, $f(-1) = 0$.

$f(-1) = 2(-1)^3 - (-1)^2 + p(-1) + q$

$10 = -2 - 1 - p + q$

$10 = -3 - p + q$

$p = q - 13$ → Equation 2

Substituting Equation 1 into Equation 2 gives:

$$q = 3(q - 13) + 63$$

$$q = 3q - 39 + 63$$

$$q - 3q = -39 + 63$$

$$-2q = 24$$

$$q = \frac{24}{-2}$$

$$q = -12$$

Substituting $q = -12$ into Equation 2 gives:

$$p = -12 - 13$$

$$= -25$$

$\therefore p = -25$ and $q = -12$

Q.E.D.

(ii) Hence, solve the equation $f(x) = 0$. [6]

$$f(x) = 2x^3 - x^2 - 25x - 12$$

We are given that $(x + 3)$ is a factor of $f(x)$.

By long division,

$$
\begin{array}{r|l}
 & 2x^2 - 7x - 4 \\
\hline
x + 3 & 2x^3 - x^2 - 25x - 12 \\
 & 2x^3 + 6x^2 \\
\hline
 & -7x^2 - 25x \\
 & -7x^2 - 21x \\
\hline
 & -4x - 12 \\
 & -4x - 12 \\
\hline
 & 0
\end{array}
$$

So, we have,

$f(x) = (x + 3)(2x^2 - 7x - 4)$

$= (x + 3)(2x + 1)(x - 4)$

Suppose $f(x) = 0$.

Then,

$(x + 3)(2x + 1)(x - 4) = 0$

Either $x + 3 = 0$ or $2x + 1 = 0$ or $x - 4 = 0$

$x = -3$ $x = -\frac{1}{2}$ $x = 4$

$\therefore x = -3, -\frac{1}{2}$ or 4

(b) Use mathematical induction to prove that $6^n - 1$ is divisible by 5 for all natural numbers n. [6]

Let $P(n)$ be the statement: $6^n - 1$ is divisible by 5 for all $n \in \mathbb{N}$.

Looking at $P(1)$:

$6^1 - 1 = 6 - 1$

$= 5$ which is divisible by 5

$\therefore P(1)$ is true.

Assume $P(k)$ is true.

That is, $6^k - 1$ is divisible by 5.

$\therefore 6^k - 1 = 5m$, where $m \in \mathbb{N}$

$\Longrightarrow 6^k = 5m + 1$

Looking at $P(k+1)$.

That is, $6^{k+1} - 1$ is divisible by 5.

We need to prove $P(k+1)$.

$6^{k+1} - 1 = 6^k \cdot 6 - 1$

$= (5m+1)(6) - 1$

$= 30m + 6 - 1$

$= 30m + 5$

$= 5(6m+1)$ which is divisible by 5

$\therefore$ If $P(k)$ is true, then $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$6^n - 1$ is divisible by 5 for all natural numbers n.

Q.E.D.

(c) (i) Given that $\boldsymbol{p}$ and $\boldsymbol{q}$ are two propositions, complete the truth table below: [4]

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{p} \to \boldsymbol{q}$	$\boldsymbol{p} \vee \boldsymbol{q}$	$\boldsymbol{p} \wedge \boldsymbol{q}$	$(\boldsymbol{p} \vee \boldsymbol{q}) \to (\boldsymbol{p} \wedge \boldsymbol{q})$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	T	F	F
F	F	T	F	F	T

(ii) State, giving a reason for your response, whether the following statements are logically equivalent:

- $\boldsymbol{p} \to \boldsymbol{q}$
- $(\boldsymbol{p} \vee \boldsymbol{q}) \to (\boldsymbol{p} \wedge \boldsymbol{q})$ [2]

$\boldsymbol{p} \to \boldsymbol{q}$ and $(\boldsymbol{p} \vee \boldsymbol{q}) \to (\boldsymbol{p} \wedge \boldsymbol{q})$ are not logically equivalent because they do not have the same truth values.

Total: 25 marks

June 2016 – Question 2 – Solution

(a) Solve the following equation for x:

$$\log_2(10 - x) + \log_2 x = 4 \qquad [6]$$

$\log_2(10 - x) + \log_2 x = 4$

$\log_2(10 - x)(x) = 4$

Converting to exponential form gives:

$(10 - x)(x) = 2^4$

$10x - x^2 = 16$

$x^2 - 10x + 16 = 0$

$(x - 8)(x - 2) = 0$

Either $x - 8 = 0$ or $x - 2 = 0$

$x = 8$ $x = 2$

$\therefore x = 2$ or $x = 8$

(b) A function f is defined by $f(x) = \frac{x+3}{x-1}$, $x \neq 1$.

Determine whether f is bijective, that is, both one-to-one and onto. [8]

Let $f(x) = \frac{x+3}{x-1}$, $x \neq 1$.

Now,

$f(a) = \frac{a+3}{a-1}$

and

$f(b) = \frac{b+3}{b-1}$

Suppose $f(a) = f(b)$.

Then, we have,

$$\frac{a+3}{a-1} = \frac{b+3}{b-1}$$

$$(a+3)(b-1) = (a-1)(b+3)$$

$$ab + 3b - a - 3 = ab + 3a - b - 3$$

$$3b - a = 3a - b$$

$$3b + b - 3a - a = 0$$

$$4b - 4a = 0$$

$$4(b-a) = 0$$

$$b - a = 0$$

$$a = b$$

Since $f(a) = f(b) \Rightarrow a = b$, then f is one-to-one.

For f to be onto, $\forall y \in Y, \exists x \in X$ such that $y = f(x)$.

Consider when $y = 1$.

There exists no value of x such that $f(x) = 1$.

Hence, f is not onto.

In order for f to be bijective, it must be one-to-one and onto. Since f is one-to-one, but not onto, then the function is not bijective.

(c) Let the roots of the equation $2x^3 - 5x^2 + 4x + 6 = 0$ be α, β and γ.

(i) State the values of $\alpha + \beta + \gamma, \alpha\beta + \alpha\gamma + \beta\gamma$ and $\alpha\beta\gamma$. [3]

We are given $2x^3 - 5x^2 + 4x + 6 = 0$

where $a = 2, b = -5, c = 4$ and $d = 6$.

Now,

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$= \frac{-(-5)}{2}$$

$$= \frac{5}{2}$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a}$$

$$= \frac{4}{2}$$

$$= 2$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$= -\frac{6}{2}$$

$$= -3$$

(ii) Hence, or otherwise, determine an equation with integer coefficients which has roots $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$. [8]

Note: $(\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

For the new roots $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$:

Sum of roots $= \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$

$$= \frac{(\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{(\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{(\alpha\beta\gamma)^2}$$

$$= \frac{(2)^2 - (2)(-3)\left(\frac{5}{2}\right)}{(-3)^2}$$

$$= \frac{4 + 15}{9}$$

$$= \frac{19}{9}$$

Pair product of roots $= \left(\frac{1}{\alpha^2}\right)\left(\frac{1}{\beta^2}\right) + \left(\frac{1}{\beta^2}\right)\left(\frac{1}{\gamma^2}\right) + \left(\frac{1}{\alpha^2}\right)\left(\frac{1}{\gamma^2}\right)$

$$= \frac{\gamma^2 + \alpha^2 + \beta^2}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{(\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\alpha\gamma+\beta\gamma)}{(\alpha\beta\gamma)^2}$$

$$= \frac{\left(\frac{5}{2}\right)^2 - 2(2)}{(-3)^2}$$

$$= \frac{\frac{25}{4} - 4}{(-3)^2}$$

$$= \frac{\frac{9}{4}}{9}$$

$$= \frac{9}{4} \div 9$$

$$= \frac{9}{4} \times \frac{1}{9}$$

$$= \frac{1}{4}$$

Product of roots $= \left(\frac{1}{\alpha^2}\right)\left(\frac{1}{\beta^2}\right)\left(\frac{1}{\gamma^2}\right)$

$$= \frac{1}{(\alpha\beta\gamma)^2}$$

$$= \frac{1}{(-3)^2}$$

$$= \frac{1}{9}$$

Hence, the new equation whose roots are $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$ is:

$x^3 - (sum\ of\ roots)x^2 + (sum\ of\ the\ product\ of\ two\ of\ the\ roots\ at\ a\ time)x - (product\ of\ the\ roots) = 0$

$$x^3 - \frac{19}{9}x^2 + \frac{1}{4}x - \frac{1}{9} = 0$$

$(\times 36)$

$$36x^3 - 76x^2 + 9x - 4 = 0$$

Total: 25 marks

June 2016 – Question 3 – Solution

(a) (i) Show that $\sec^2\theta = \frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta - \sin\theta}$. [4]

Required to show that $\sec^2\theta = \frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta - \sin\theta}$.

Taking R.H.S:

$$\frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta - \sin\theta} = \frac{\frac{1}{\sin\theta}}{\frac{1}{\sin\theta} - \sin\theta}$$

$$= \frac{\frac{1}{\sin\theta}}{\frac{1-\sin^2\theta}{\sin\theta}}$$

$$= \frac{1}{\sin\theta} \div \frac{1-\sin^2\theta}{\sin\theta}$$

$$= \frac{1}{\sin\theta} \times \frac{\sin\theta}{1-\sin^2\theta}$$

$$= \frac{1}{1-\sin^2\theta}$$

$$= \frac{1}{\cos^2\theta}$$

$$= \sec^2\theta$$

$$= \text{L.H.S.}$$

$$\therefore \sec^2\theta = \frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta - \sin\theta}$$

Q.E.D.

(ii) Hence, or otherwise, solve the equation $\frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta - \sin\theta} = \frac{4}{3}$ for $0 \le \theta \le 2\pi$. [5]

$$\frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta - \sin\theta} = \frac{4}{3}$$

$$\sec^2\theta = \frac{4}{3}$$

$$\frac{1}{\cos^2\theta} = \frac{4}{3}$$

$$\cos^2\theta = \frac{3}{4}$$

$$\cos\theta = \pm\frac{\sqrt{3}}{2}$$

		$\frac{\pi}{6}$
$-$	$+$	
$-$	$+$	

$$\theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ for $0 \le \theta \le 2\pi$.

(b) (i) Express the function $f(\theta) = \sin\theta + \cos\theta$ in the form $r\sin(\theta + \alpha)$, where

$r > 0$ and $0 \le \alpha \le \frac{\pi}{2}$. [5]

Let $\sin\theta + \cos\theta = r\sin(\theta + \alpha)$

$$\sin\theta + \cos\theta = r\sin\theta\cos\alpha + r\cos\theta\sin\alpha$$

Equating coefficients gives:

$r\cos\alpha = 1$ $\rightarrow$ Equation 1

$r\sin\alpha = 1$ $\rightarrow$ Equation 2

Equation 2 ÷ Equation 1 gives:

$$\frac{r\sin\alpha}{r\cos\alpha} = \frac{1}{1}$$

$$\tan\alpha = 1$$

$$\alpha = \tan^{-1}(1)$$

$$\alpha = \frac{\pi}{4}$$

Now,

$$r = \sqrt{1^2 + 1^2}$$

$$= \sqrt{2}$$

$$\therefore \sin\theta + \cos\theta = \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right)$$

(ii) Hence, find the maximum value of f and the **smallest** non-negative value of θ at which it occurs. [5]

$$f(\theta) = \sin\theta + \cos\theta$$

$$f(\theta) = \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right)$$

Now,

$$-1 \leq \sin\theta \leq 1$$

$$-1 \leq \sin\left(\theta + \frac{\pi}{4}\right) \leq 1$$

$$-\sqrt{2} \leq \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right) \leq \sqrt{2}$$

Hence, the maximum value of $f(\theta)$ is $\sqrt{2}$.

So,

$$\sqrt{2}\sin\left(\theta+\frac{\pi}{4}\right)=\sqrt{2}$$

$$\sin\left(\theta+\frac{\pi}{4}\right)=1$$

$$\theta+\frac{\pi}{4}=\frac{\pi}{2}$$

$$\theta=\frac{\pi}{2}-\frac{\pi}{4}$$

$$\theta=\frac{\pi}{4}$$

$\therefore$ The smallest non-negative value of θ at which it occurs is $\theta=\frac{\pi}{4}$.

(c) Prove that

$$\tan(A+B+C)=\frac{\tan A+\tan B+\tan C-\tan A\tan B\tan C}{1-\tan A\tan B-\tan A\tan C-\tan B\tan C}$$ [6]

Required to prove that $\tan(A+B+C)=\frac{\tan A+\tan B+\tan C-\tan A\tan B\tan C}{1-\tan A\tan B-\tan A\tan C-\tan B\tan C}$.

Taking L.H.S:

$$\tan(A+B+C)=\tan\big(A+(B+C)\big)$$

$$\tan(A+B+C)=\frac{\tan A+\tan(B+C)}{1-\tan A\tan(B+C)} \qquad \rightarrow \text{Equation 1}$$

Consider the numerator of Equation 1:

$$\tan A + \tan(B + C) = \tan A + \frac{\tan B + \tan C}{1 - \tan B \tan C}$$

$$= \frac{\tan A(1 - \tan B \tan C) + \tan B + \tan C}{1 - \tan B \tan C}$$

$$= \frac{\tan A - \tan A \tan B \tan C + \tan B + \tan C}{1 - \tan B \tan C}$$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan B \tan C}$$

Consider the denominator of Equation 1:

$$1 - \tan A \tan(B + C) = 1 - \tan A \left(\frac{\tan B + \tan C}{1 - \tan B \tan C}\right)$$

$$= 1 - \frac{\tan A \tan B + \tan A \tan C}{1 - \tan B \tan C}$$

$$= \frac{1 - \tan B \tan C - \tan A \tan B + \tan A \tan C}{1 - \tan B \tan C}$$

So, we have,

$$\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan B \tan C} \div \frac{1 - \tan B \tan C - \tan A \tan B + \tan A \tan C}{1 - \tan B \tan C}$$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan B \tan C} \times \frac{1 - \tan B \tan C}{1 - \tan B \tan C - \tan A \tan B + \tan A \tan C}$$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan B \tan C - \tan A \tan B + \tan A \tan C}$$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan A \tan C - \tan B \tan C}$$

$$= \text{R.H.S.}$$

$$\therefore \tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan A \tan C - \tan B \tan C}$$

Q.E.D.

Total: 25 marks

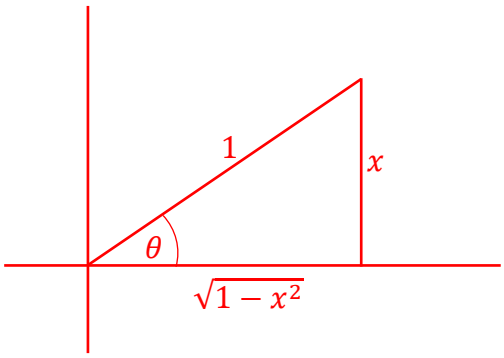
June 2016 – Question 4 – Solution

(a) (i) Given that $\sin\theta = x$, show that $\tan\theta = \frac{x}{\sqrt{1-x^2}}$, where $0 < \theta < \frac{\pi}{2}$. [3]

Now,

$\sin\theta = x$

$\sin\theta = \frac{x}{1} = \frac{opp}{hyp}$



Using Pythagoras' Theorem,

$$a^2 + b^2 = c^2$$

$$(x)^2 + b^2 = 1^2$$

$$x^2 + b^2 = 1$$

$$b^2 = 1 - x^2$$

$$b = \sqrt{1 - x^2}$$

Now,

$\tan\theta = \frac{opp}{adj}$

$= \frac{x}{\sqrt{1-x^2}}$

$\therefore \tan\theta = \frac{x}{\sqrt{1-x^2}}$

Q.E.D.

(ii) Hence, or otherwise, determine the Cartesian equation of the curve defined parametrically by $y = \tan 2t$ and $x = \sin t$ for $0 < t < \frac{\pi}{2}$. [5]

$y = \tan 2t$

$y = \frac{2\tan t}{1-\tan^2 t}$

Substituting $\tan t = \frac{x}{\sqrt{1-x^2}}$ gives:

$$y = \frac{2\left(\frac{x}{\sqrt{1-x^2}}\right)}{1-\left(\frac{x}{\sqrt{1-x^2}}\right)^2}$$

$$= \frac{\left(\frac{2x}{\sqrt{1-x^2}}\right)}{1-\frac{x^2}{1-x^2}}$$

$$= \left(\frac{2x}{\sqrt{1-x^2}}\right) \div \left(1 - \frac{x^2}{1-x^2}\right)$$

$$= \left(\frac{2x}{\sqrt{1-x^2}}\right) \div \left(\frac{1-x^2-x^2}{1-x^2}\right)$$

$$= \frac{2x}{\sqrt{1-x^2}} \div \frac{1-2x^2}{1-x^2}$$

$$= \frac{2x}{\sqrt{1-x^2}} \times \frac{1-x^2}{1-2x^2}$$

$$= \frac{2x\sqrt{1-x^2}}{1-2x^2}$$

$\therefore$ The Cartesian equation of the curve is $y = \frac{2x\sqrt{1-x^2}}{1-2x^2}$.

(b) Let $\boldsymbol{u} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$ and $\boldsymbol{v} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$ be two position vectors in $\mathbb{R}^3$.

(i) Calculate the lengths of $\boldsymbol{u}$ and $\boldsymbol{v}$ respectively. [3]

We have,

$$\boldsymbol{u} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$$

$$|\boldsymbol{u}| = \sqrt{(1)^2 + (-3)^2 + (2)^2}$$

$$= \sqrt{1 + 9 + 4}$$

$$= \sqrt{14} \text{ units}$$

Now,

$$\boldsymbol{v} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$$

$$|\boldsymbol{v}| = \sqrt{(2)^2 + (1)^2 + (5)^2}$$

$$= \sqrt{4 + 1 + 25}$$

$$= \sqrt{30} \text{ units}$$

$\therefore$ The lengths of $\boldsymbol{u}$ and $\boldsymbol{v}$ are $\sqrt{14}$ units and $\sqrt{30}$ units respectively.

(ii) Find $\cos\theta$ where θ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\mathbb{R}^3$. [4]

Required to find $\cos\theta$ where θ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\mathbb{R}^3$.

$\cos\theta = \frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{u}||\boldsymbol{v}|}$, where θ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\mathbb{R}^3$.

Now,

$$\boldsymbol{u}\cdot\boldsymbol{v} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$$

$$= (1)(2) + (-3)(1) + (2)(5)$$

$$= 2 - 3 + 10$$

$$= 9$$

So, we have,

$$\cos\theta = \frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{u}||\boldsymbol{v}|}$$

$$= \frac{3}{\sqrt{14}\sqrt{30}}$$

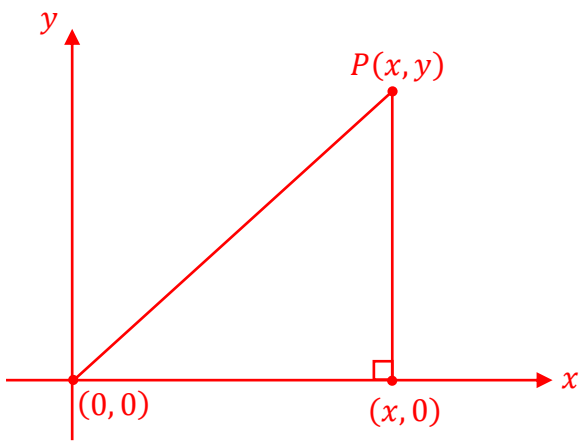
$$= \frac{9}{\sqrt{420}}$$

$$\therefore \cos\theta = \frac{9}{\sqrt{420}}$$

(c) A point, $P(x, y)$, moves such that its distance from the x-axis is half its distance from the origin.

Determine the Cartesian equation of the locus of P. [5]

A point, $P(x, y)$, moves such that its distance from the x-axis is half its distance from the origin.



Consider points $(0, 0)$ and $P(x, y)$.

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(x - 0)^2 + (y - 0)^2}$$

$$= \sqrt{x^2 + y^2}$$

Consider points $P(x, y)$ and $(x, 0)$.

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(x - x)^2 + (y - 0)^2}$$

$$= \sqrt{y^2}$$

$$= y$$

So, we have,

$\frac{1}{2}\sqrt{x^2+y^2} = y$

$\sqrt{x^2+y^2} = 2y$

Squaring both sides:

$$x^2+y^2 = (2y)^2$$

$$x^2+y^2 = 4y^2$$

$x^2+y^2-4y^2 = 0$

$$x^2-3y^2 = 0$$

$\therefore$ The Cartesian equation of the locus of P is: $x^2-3y^2=0$

(d) The line L has the equation $2x+y+3=0$ and the circle C has the equation

$x^2+y^2=9$. Determine the points of intersection of the circle C and the line L. [5]

Required to determine the points of intersection of the circle C and the line L.

$2x+y+3=0$ $\rightarrow$ Equation 1

$x^2+y^2=9$ $\rightarrow$ Equation 2

Rearranging Equation 1 gives:

$y=-2x-3$ $\rightarrow$ Equation 3

Substituting Equation 3 into Equation 2 gives:

$$x^2 + (-2x - 3)^2 = 9$$

$$x^2 + [(-1)(2x + 3)]^2 = 9$$

$$x^2 + (2x + 3)^2 = 9$$

$$x^2 + 4x^2 + 12x + 9 = 9$$

$$5x^2 + 12x = 0$$

$$x(5x + 12) = 0$$

Either $x = 0$ or $5x + 12 = 0$

$$x = -\frac{12}{5}$$

When $x = 0$,

$y = -2(0) - 3$

$= 0 - 3$

$= -3$

When $x = -\frac{12}{5}$,

$y = -2\left(-\frac{12}{5}\right) - 3$

$= \frac{24}{5} - 3$

$= \frac{9}{5}$

$\therefore$ The points of intersection are $(0, -3)$ and $\left(-\frac{12}{5}, \frac{9}{5}\right)$.

Total: 25 marks

June 2016 – Question 5 – Solution

(a) Use an appropriate substitution to find $\int (x+1)^{\frac{1}{3}}\,dx$. [4]

Required to use an appropriate substitution to find $\int (x+1)^{\frac{1}{3}}\,dx$.

Let $u = x + 1$

$$\frac{du}{dx} = 1$$

$$dx = du$$

Now,

$$\int (x+1)^{\frac{1}{3}}\,dx = \int u^{\frac{1}{3}}\,du$$

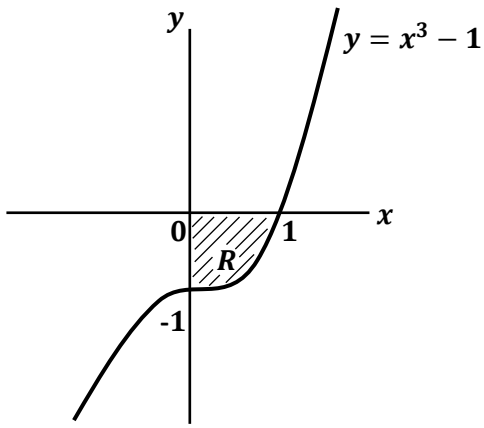
$$= \frac{u^{\frac{4}{3}}}{\frac{4}{3}} + c$$

$$= \frac{3}{4}u^{\frac{4}{3}} + c$$

$$= \frac{3}{4}(x+1)^{\frac{4}{3}} + c$$

$$\therefore \int (x+1)^{\frac{1}{3}}\,dx = \frac{3}{4}(x+1)^{\frac{4}{3}} + c$$

(b) The diagram below represents the finite region R which is enclosed by the curve $y = x^3 - 1$ and the lines $x = 0$ and $y = 0$.



Calculate the volume of the solid that results from rotating R about the y-axis. [5]

$y = x^3 - 1$

$x^3 = y + 1$

$x = (y+1)^{\frac{1}{3}}$

$x^2 = (y+1)^{\frac{2}{3}}$

Volume about the y-axis $= \pi \int_a^b x^2\, dy$

$= \pi \int_{-1}^{0} (y+1)^{\frac{2}{3}}\, dy$

$= \pi \left[\frac{3}{5}(y+1)^{\frac{5}{3}}\right]_{-1}^{0}$

$= \frac{3\pi}{5}\left[(y+1)^{\frac{5}{3}}\right]_{-1}^{0}$

$= \frac{3\pi}{5}\left[(0+1)^{\frac{5}{3}} - (-1+1)^{\frac{5}{3}}\right]$

$= \frac{3\pi}{5}\left[(1)^{\frac{5}{3}} - (0)^{\frac{5}{3}}\right]$

$= \frac{3\pi}{5}$ units^3

(c) Given that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx \quad a > 0$, show that

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}\,dx = \frac{1}{2}.$$ [6]

Given that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ for $a > 0$,

Then for $a = 1$ and $f(x) = \frac{e^x}{e^x+e^{1-x}}$, consider:

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \int_0^1 \frac{e^x+e^{1-x}-e^{1-x}}{e^x+e^{1-x}}dx$$

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \int_0^1 \left(\frac{e^x+e^{1-x}}{e^x+e^{1-x}} - \frac{e^{1-x}}{e^x+e^{1-x}}\right)dx$$

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \int_0^1 \left(1 - \frac{e^{1-x}}{e^x+e^{1-x}}\right)dx$$

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \int_0^1 1\,dx - \int_0^1 \frac{e^{1-x}}{e^{1-x}+e^x}\,dx$$

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \int_0^1 1\,dx - \int_0^1 \frac{e^x}{e^x+e^{1-x}}dx$$

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx + \int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \int_0^1 1\,dx$$

$$2\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = [x]_0^1$$

$$2\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = (1-0)$$

$$2\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = 1$$

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \frac{1}{2}$$

Recall:

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$$\int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \int_0^1 \frac{e^{1-x}}{e^{1-x}+e^x}\,dx$$

$$\therefore \int_0^1 \frac{e^x}{e^x+e^{1-x}}dx = \frac{1}{2}$$

Q.E.D.

(d) An initial population of 10 000 bacteria grow exponentially at a rate of 2% per hour, where $y = f(t)$ is the number of bacteria present t hours later.

(i) Solve an appropriate differential equation to show that the number of bacteria present at any time can be modelled by the equation $y = 10\,000e^{0.02t}$. [7]

Let $\frac{dy}{dt}$ be the rate of growth of bacteria.

Then,

$$\frac{dy}{dt} = 0.02y$$

Separating variables gives:

$$\frac{1}{y}\,dy = 0.02\,dt$$

Integrating both sides gives:

$$\int \frac{1}{y}\,dy = \int 0.02\,dt$$

$$\ln y = 0.02t + c$$

$$y = e^{0.02t+c}$$

When $t = 0, y = 10\,000$,

$$10\,000 = e^{0.02(0)+c}$$

$$10\,000 = e^{c}$$

$$\ln 10\,000 = c \ln e$$

$$c = \ln 10\,000$$

Substituting $c = \ln 10\ 000$ into $y = e^{0.02t+c}$ gives:

$$y = e^{0.02t+\ln 10\ 000}$$

$$= e^{0.02t} \cdot e^{\ln 10\ 000}$$

$$= e^{0.02t} \cdot 10\ 000$$

$$= 10\ 000e^{0.02t}$$

∴ The number of bacteria present at any time can be modelled by the equation:

$$y = 10\ 000e^{0.02t}$$

Q.E.D.

(ii) Determine the time required for the bacteria population to double in size. [3]

Initial population $= 10\ 000$

New population $= 10\ 000 \times 2$

$= 20\ 000$

Now,

$$20\ 000 = 10\ 000e^{0.02t}$$

$$e^{0.02t} = \frac{20\ 000}{10\ 000}$$

$$e^{0.02t} = 2$$

Taking ln on both sides:

$$\ln e^{0.02t} = \ln 2$$

$$0.02t \ln e = \ln 2$$

$$0.02t = \ln 2$$

$$t = \frac{\ln 2}{0.02}$$

$$t = 34.7 \text{ hours} \quad \text{(to 3 significant figures)}$$

∴ The time required for the bacteria population to double in size is 34.7 hours.

Total: 25 marks

June 2016 – Question 6 – Solution

(a) Find the equation of the tangent to the curve $f(x) = 2x^3 + 5x^2 - x + 12$ at the point where $x = 3$. [4]

$f(x) = 2x^3 + 5x^2 - x + 12$

When $x = 3$,

$f(3) = 2(3)^3 + 5(3)^2 - (3) + 12$

$= 54 + 45 - 3 + 12$

$= 108$

Now,

$f'(x) = 6x^2 + 10x - 1$

When $x = 3$,

$f'(3) = 6(3)^2 + 10(3) - 1$

$= 54 + 30 - 1$

$= 83$

Substituting $m = 83$ and point $(3, 108)$ into $y - y_1 = m(x - x_1)$ gives:

$y - 108 = 83(x - 3)$

$y = 83x - 249 + 108$

$y = 83x - 141$

$\therefore$ The equation of the tangent to the curve is: $y = 83x - 141$

(b) A function f is defined on $\mathbb{R}$ as

$$f(x) = \begin{cases} x^2 + 2x + 3 & x \leq 0 \\ ax + b & x > 0 \end{cases}$$

(i) Calculate the $\lim_{x \to 0^-} f(x)$ and $\lim_{x \to 0^+} f(x)$. [4]

$$f(x) = \begin{cases} x^2 + 2x + 3 & x \leq 0 \\ ax + b & x > 0 \end{cases}$$

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (x^2 + 2x + 3)$$

$$= (0)^2 + 2(0) + 3$$

$$= 0 + 0 + 3$$

$$= 3$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (ax + b)$$

$$= a(0) + b$$

$$= 0 + b$$

$$= b$$

(ii) Hence, determine the values of a and b such that $f(x)$ is continuous at $x = 0$. [5]

For $f(x)$ to be continuous at $x = 0$, $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x)$.

So, we have, $b = 3$.

Now,

$f(x) = ax + 3$

$\therefore$ The value of $b = 3$ and a could assume any value, that is, $a \in \mathbb{R}$.

(iii) If the value of $b = 3$, determine a such that $f'(0) = \lim_{t \to 0} \frac{f(0+t) - f(0)}{t}$. [6]

Since $b = 3$, then $f(x) = \begin{cases} x^2 + 2x + 3 & x \leq 0 \\ ax + 3 & x > 0 \end{cases}$

$$f'(0) = \lim_{t \to 0} \frac{f(0+t) - f(0)}{t}$$

$$= \lim_{t \to 0} \frac{a(0+t) + 3 - [(0)^2 + 2(0) + 3]}{t}$$

$$= \lim_{t \to 0} \frac{0 + at + 3 - (0 + 0 + 3)}{t}$$

$$= \lim_{t \to 0} \frac{at + 3 - 3}{t}$$

$$= \lim_{t \to 0} \frac{at}{t}$$

$$= \lim_{t \to 0} a$$

$$= a$$

Now, $f'(x) = \begin{cases} 2x+2 & x \leq 0 \\ a & x > 0 \end{cases}$.

$f'(0) = 2(0) + 2$

$= 2$

Since $f'(0) = a$ and $f'(0) = 2$, then $a = 2$.

$\therefore$ The value of $a = 2$.

(c) Use first principles to differentiate $f(x) = \sqrt{x}$ with respect to x. [6]

Let $f(x) = \sqrt{x}$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$

$$= \lim_{h \to 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \to 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$= \frac{1}{\sqrt{x+0} + \sqrt{x}}$$

$$= \frac{1}{\sqrt{x} + \sqrt{x}}$$

$$= \frac{1}{2\sqrt{x}}$$

$$\therefore f'(x) = \frac{1}{2\sqrt{x}}$$

Total: 25 marks